Generalized Pandrosion Residuals

Residual profile of an iterative geometric construction for p-th roots of any positive real number

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Abstract

We revisit Pandrosion's classical geometric construction, reported by Pappus, for approximating the cube root of two and generalize it to the computation of pth roots of any positive real x by inserting proportional means using Thales' theorem. In our formulation the construction reduces to a single parameter $s_n \in (0,1)$ together with an output v_n converging to $x^{1/p}$. After a normalization the iteration takes a universal form involving the geometric sum

$$S_p(s) = 1 + s + \cdots + s^{p-1}.$$

We identify the unique fixed point $s^* = x^{-1/p}$, show that the convergence is *linear*, and derive an explicit closed-form expression for the associated contraction ratio

$$\lambda_{p,x} = \frac{(\alpha - 1)\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}, \qquad \alpha = x^{1/p}.$$

This ratio depends only on the target α and provides a concise "convergence signature" for Pandrosion's method. Building on this formula we obtain non-asymptotic error bounds, prove that $\lambda_{p,x}$ is monotonic in both the order p and the argument x, extend the construction to the case $0 < x < 1$ (where convergence becomes oscillatory), and determine an optimal starting point that minimizes the initial error. We further show that Steffensen's acceleration applied to the Pandrosion iteration yields *quadratic* convergence, and we propose a simple scaling strategy that exploits the monotonicity of $\lambda_{p,x}$ to reduce the computational cost for large x. Taken together, these results situate an ancient geometric procedure within modern fixed-point theory and suggest derivative-free root-finding algorithms inspired by Pandrosion.

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1 Introduction

The Greek problem of doubling the cube consists of constructing a segment of length $\sqrt[3]{2}$ from a unit segment. It is equivalent to finding two proportional means m_1, m_2 between 1 and 2, that is,

$$\frac{1}{m_1} = \frac{m_1}{m_2} = \frac{m_2}{2},$$

which yields $m_1 = \sqrt[3]{2}$ and $m_2 = \sqrt[3]{4}$. Pandrosion of Alexandria proposed an *approximate* geometric method, reported by Pappus in the *Collectio*, Book III [3]: by successive parallels

(Thales' theorem), she constructs proportional ratios that approach these means without ever reaching them exactly in a finite number of steps.

This work starts from a fundamental observation:

> *Mathematically the target number is exact, but any approximation procedure yields a measurable residual profile — a footprint intrinsic to the chosen method.*

At each finite step of Pandrosion's construction, there remains a gap $\varepsilon_n = |v_n - \alpha|$ between the geometric approximation and the ideal value. The sequence (ε_n) is the **residual profile** of the method. Our goal is to compute this profile exactly for any p-th root $x^{1/p}$, and to identify the contraction ratio that characterizes Pandrosion's method.

To our knowledge, the explicit expression for $\lambda_{p,x}$ in equation (5) is not recorded in standard numerical analysis textbooks (such as Stoer–Bulirsch [5] or the third edition of *Numerical Recipes* [6]) nor in the historical literature on cube duplication. We therefore include its derivation here for completeness.

Outline. Section 2 recalls Pandrosion's construction for $\sqrt[3]{2}$ and analyzes its residual profile. Section 4 states the generalization to any p, x. Section 5 computes the contraction ratio $\lambda_{p,x}$ and provides closed-form expressions. Section 7 presents numerical tables. Section 8 compares with bisection, the secant method, and Newton's method. Section 9 establishes functional properties of $\lambda_{p,x}$. Section 10 provides non-asymptotic error bounds. Section 11 extends the theory to $x < 1$. Section 12 determines the optimal starting point. Section 13 shows that Steffensen's acceleration of Pandrosion achieves quadratic convergence. Section 14 exploits the monotonicity of $\lambda_{p,x}$ to optimize the method for large x via scaling. Section 15 concludes.

2 Pandrosion's construction for $\sqrt[3]{2}$

2.1 The two sequences

The geometric construction produces two sequences as a function of an initial parameter u_0:

$$u_{n+1} = \frac{u_n}{2 - 2\left(\dfrac{4-u_n}{4}\right)^3}, \qquad v_n = 2\left(\frac{4-u_n}{4}\right)^2, \qquad u_0 = 0.5. \tag{1}$$

The sequence (u_n) is the *internal parameter* of the construction (a length in the Thales diagram). The sequence (v_n) is the *readout*: the approximation of $\sqrt[3]{2}$ produced at step n.

Proposition 2.1. $\lim_{n\to\infty} v_n = \sqrt[3]{2}$.

2.2 Normalized substitution

Setting $s_n = (4 - u_n)/4$, so that $s_n \in (0, 1)$, we obtain:

- $u_n = 4(1 - s_n)$,
- $v_n = 2s_n^2$,
- $u_{n+1} = \dfrac{4(1-s_n)}{2(1-s_n^3)} = \dfrac{2}{1 + s_n + s_n^2}$.

The iteration on s_n then becomes:

$$s_{n+1} = 1 - \frac{u_{n+1}}{4} = \frac{2s_n^2 + 2s_n + 1}{2s_n^2 + 2s_n + 2}. \tag{2}$$

2.3 Fixed point

The fixed point s^* satisfies $s^* = (2s^{*2}+2s^*+1)/(2s^{*2}+2s^*+2)$, which gives $2(1-s^*)(1+s^*+s^{*2}) = 1$, i.e. $2(1-s^{*3}) = 1$, hence $s^{*3} = 1/2$, so:

$$s^* = 2^{-1/3} = \frac{1}{\sqrt[3]{2}}.$$

The output at the fixed point is $v^* = 2s^{*2} = 2 \cdot 2^{-2/3} = 2^{1/3} = \sqrt[3]{2}$. ✓

2.4 Residual profile: linear convergence

Let $h(s) = (2s^2 + 2s + 1)/(2s^2 + 2s + 2)$. The derivative at s^* gives the convergence rate:

$$h'(s^*) = \frac{4s^* + 2}{(2s^{*2} + 2s^* + 2)^2} = \frac{2(2s^* + 1)}{4(s^{*2} + s^* + 1)^2} = \frac{2s^* + 1}{2(s^{*2} + s^* + 1)^2}.$$

Substituting $s^* = 2^{-1/3}$ and using the closed form (7):

$$\lambda_{3,2} = \frac{(\sqrt[3]{2} - 1)(\sqrt[3]{2} + 2)}{\sqrt[3]{4} + \sqrt[3]{2} + 1} \approx 0.2202.$$

Theorem 2.1 (Linear residual law for $p = 3$, $x = 2$)**.** Under the above hypotheses, for any s_0 sufficiently close to s^*:

$$\varepsilon_{n+1}^{(s)} \approx \lambda_{3,2}\, \varepsilon_n^{(s)}, \qquad \lambda_{3,2} \approx 0.2202.$$

Since $v_n - \sqrt[3]{2} \approx (p-1)\alpha^2\,(s_n - s^*)$ and $(p-1)\alpha^2 = 2 \cdot \sqrt[3]{4}$, the output residual of v_n converges at the same linear rate.

n	s_n	v_n	$\varepsilon_n = \lvert v_n - \sqrt[3]{2}\rvert$	$\varepsilon_{n+1}/\varepsilon_n$
0	0.8750	1.5312	2.713×10^{-1}	—
1	0.8107	1.3143	5.44×10^{-2}	0.200
2	0.7974	1.2717	1.17×10^{-2}	0.216
3	0.7945	1.2625	2.58×10^{-3}	0.219
4	0.7939	1.2605	5.67×10^{-4}	0.220
∞	$2^{-1/3}$	$\sqrt[3]{2}$	0	$\to \lambda_{3,2}$

Table 1: Numerical residual profile for $p = 3$, $x = 2$ ($u_0 = 0.5$, i.e. $s_0 = 0.875$). The ratio converges to $\lambda_{3,2} \approx 0.220$.

3 Geometric origin: why S_p arises naturally

3.1 The construction by successive parallels

Pandrosion's construction relies on Thales' theorem applied to a rectangle. Consider a rectangle with sides H and W equipped with two diagonals (Figure 1). A horizontal line at height u_n from the bottom intersects the two diagonals and defines:

- the parameter $s_n = (H - u_n)/H$ (normalized position from the top),
- the readout length $v_n = x \cdot s_n^{p-1}$ (the output of the construction).

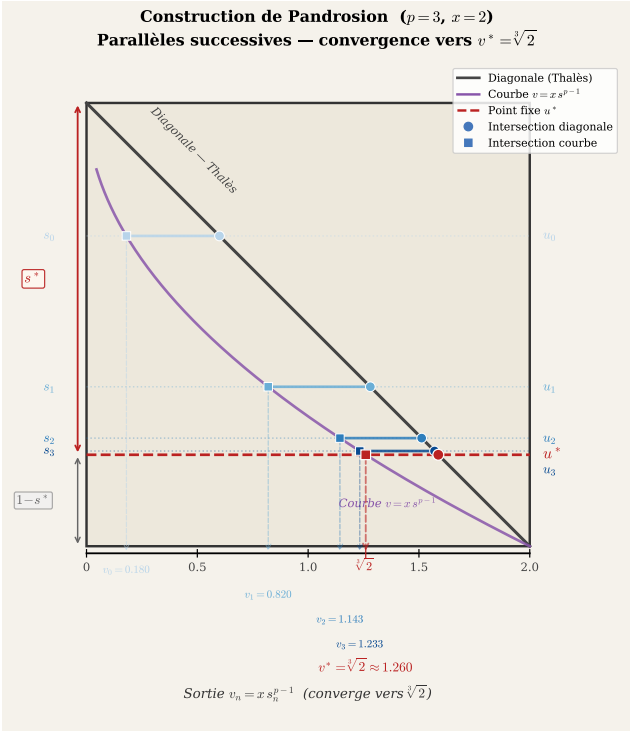

Figure 1: Pandrosion's geometric construction for $p = 3$, $x = 2$. The blue horizontal lines (steps $n = 0, 1, 2, 3$) converge toward the dashed red line, which corresponds to $v^* = \sqrt[3]{2}$.

3.2 Where does the polynomial S_p come from?

The fundamental idea is as follows. We seek to insert $p-1$ proportional means between 1 and x, that is, to construct the ideal geometric sequence $1, \alpha, \alpha^2, \ldots, \alpha^{p-1}, x$ where $\alpha = x^{1/p}$. If $s_n \approx \alpha^{-1}$ is the current approximation of the inverse ratio, the sum of the approximate geometric sequence is:

$$S_p(s_n) = 1 + s_n + s_n^2 + \cdots + s_n^{p-1}.$$

The ideal sequence satisfies $x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1$, which rewrites as $x(1 - s^{*p}) = x - 1$. Pandrosion's correction adjusts s_n so as to reduce the gap between $S_p(s_n)$ and its ideal value via the formula:

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)}.$$

Geometrically, the ratio $(x-1)/(x \cdot S_p(s_n))$ measures the *deviation* of the current proportional chain from the perfect geometric chain, and subtraction from 1 produces the corrected ratio. The polynomial S_p is thus the analytical transcription of the sum of the p segments constructed via Thales' theorem.

4 Generalization: Pandrosion's construction for $x^{1/p}$

4.1 The universal iteration

Let $x > 0$ and $p \geq 2$ be an integer. Set $\alpha = x^{1/p}$. The generalized Pandrosion geometric construction is defined as follows.

Definition 4.1 (Generalized Pandrosion iteration)**.** We define the sequence $(s_n)_{n \geq 0}$ in $(0, 1)$ by

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k = \frac{1-s^p}{1-s}, \tag{3}$$

and the output sequence by

$$v_n = x \cdot s_n^{p-1}. \tag{4}$$

We call $\varepsilon_n = |v_n - \alpha|$ the **Pandrosion residual** at step n.

Remark 4.1. For $p = 3$, $x = 2$, we recover exactly the iteration (2). Indeed:

$$1 - \frac{1}{2(1+s+s^2)} = \frac{2s^2+2s+1}{2s^2+2s+2}. \checkmark$$

4.2 Geometric interpretation

The iteration (3) analytically translates the Thales operation: at each step, $S_p(s_n)$ measures the "geometric sum" of the constructed segments, and the correction $(x-1)/(x \cdot S_p(s_n))$ adjusts the position proportionally to the gap with the ideal geometric sequence $1, \alpha, \alpha^2, \ldots, \alpha^{p-1}, x$.

4.3 Fixed point

Theorem 4.1 (Fixed point)**.** The unique fixed point of the iteration (3) in $(0,1)$ is $s^* = x^{-1/p} = \alpha^{-1}$. The corresponding output is $v^* = x \cdot (x^{-1/p})^{p-1} = x^{1/p} = \alpha$.

Proof. At a fixed point s^*, we have $1 - (x-1)/(x \cdot S_p(s^*)) = s^*$, i.e.

$$x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1.$$

Since $S_p(s)(1-s) = 1 - s^p$, this gives $x(1 - s^{*p}) = x - 1$, hence $s^{*p} = 1/x$, i.e. $s^* = x^{-1/p}$. The formula for v^* follows immediately. □

5 The contraction ratio $\lambda_{p,x}$

5.1 General computation

Theorem 5.1 (Pandrosion contraction ratio)**.** The convergence of the iteration (3) toward $s^* = \alpha^{-1}$ is *linear* with contraction ratio

$$\boxed{\lambda_{p,x} = \frac{(\alpha-1)\displaystyle\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\displaystyle\sum_{k=0}^{p-1} \alpha^k}}, \qquad \alpha = x^{1/p}. \tag{5}$$

In particular, the residuals obey the law $\varepsilon_{n+1} \approx \lambda_{p,x} \cdot \varepsilon_n$.

Proof. Let $h(s) = 1 - (x-1)/(x \cdot S_p(s))$. Then

$$h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2},$$

where $S_p'(s) = \sum_{k=1}^{p-1} k s^{k-1}$.

At the fixed point $s^* = \alpha^{-1}$:

$$S_p(s^*) = \frac{1-\alpha^{-p}}{1-\alpha^{-1}} = \frac{\alpha(x-1)}{x(\alpha-1)},$$

$$S_p'(s^*) = \sum_{k=1}^{p-1} k\,\alpha^{-(k-1)} = \alpha \sum_{k=1}^{p-1} k\,\alpha^{-k}.$$

Substituting:

$$h'(s^*) = \frac{x-1}{x} \cdot \frac{\alpha \sum_{k=1}^{p-1} k\,\alpha^{-k}}{\left[\dfrac{\alpha(x-1)}{x(\alpha-1)}\right]^2} = \frac{x(\alpha-1)^2}{\alpha(x-1)} \sum_{k=1}^{p-1} k\,\alpha^{-k}.$$

Multiplying numerator and denominator by α^{p-1}, and using $x = \alpha^p$:

$$h'(s^*) = \frac{(\alpha-1)^2\,\alpha^{p-1} \sum_{k=1}^{p-1} k\,\alpha^{-k}}{\alpha^p - 1} = \frac{(\alpha-1) \sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k},$$

where we used $\alpha^p - 1 = (\alpha-1)\sum_{k=0}^{p-1} \alpha^k$. This establishes (5). □

Remark 5.1. The contraction ratio $\lambda_{p,x}$ is entirely determined by $\alpha = x^{1/p}$: the target itself dictates the rate at which Pandrosion's method approaches it. This is the **convergence signature** of the method.

5.2 Closed forms for $p = 2$ and $p = 3$

For $p = 2$: $\sum_{k=1}^{1} k\,\alpha^0 = 1$, $\sum_{k=0}^{1} \alpha^k = 1 + \alpha$.

$$\lambda_{2,x} = \frac{\alpha - 1}{\alpha + 1} = \frac{\sqrt{x} - 1}{\sqrt{x} + 1}. \tag{6}$$

For $p = 3$: $\sum_{k=1}^{2} k\,\alpha^{2-k} = \alpha + 2$, $\sum_{k=0}^{2} \alpha^k = \alpha^2 + \alpha + 1$.

$$\lambda_{3,x} = \frac{(\alpha-1)(\alpha+2)}{\alpha^2 + \alpha + 1} = \frac{(\sqrt[3]{x} - 1)(\sqrt[3]{x} + 2)}{\sqrt[3]{x^2} + \sqrt[3]{x} + 1}. \tag{7}$$

For $p = 4$: $\sum_{k=1}^{3} k\,\alpha^{3-k} = \alpha^2 + 2\alpha + 3$, $\sum_{k=0}^{3} \alpha^k = \alpha^3 + \alpha^2 + \alpha + 1$.

$$\lambda_{4,x} = \frac{(\alpha-1)(\alpha^2 + 2\alpha + 3)}{\alpha^3 + \alpha^2 + \alpha + 1}. \tag{8}$$

Remark 5.2 (Asymptotic behavior in x)**.** For fixed $p = 2$ and $x \to \infty$: $\lambda_{2,x} = (\sqrt{x} - 1)/(\sqrt{x} + 1) \to 1^-$. The larger x, the slower the convergence. For $x \to 1^+$: $\lambda_{p,x} \to 0$, convergence is very fast near 1.

5.3 Basin of attraction

Theorem 5.2 (Global convergence)**.** For any $x > 1$ and any $p \geq 2$, the iteration (3) converges to s^* for all $s_0 \in (0, 1)$. Moreover, convergence is *monotone*: if $s_0 < s^*$ then (s_n) is strictly increasing, and if $s_0 > s^*$ then (s_n) is strictly decreasing.

Proof. Set $h(s) = 1 - (x-1)/(x \cdot S_p(s))$.

Step 1: h is strictly increasing. We have $h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2}$, where $S_p'(s) = \sum_{k=1}^{p-1} k s^{k-1} > 0$ for $s > 0$.

Step 2: h maps $(0, 1)$ into itself. For $s \in (0, 1)$: $S_p(s) > 1$ (since the first term is 1 and the remaining terms are positive), so $h(s) = 1 - (x-1)/(x \cdot S_p(s)) < 1 - (x-1)/(x \cdot p) > 0$. More precisely, $h(0^+) = 1 - (x-1)/x = 1/x > 0$ and $h(1^-) = 1 - (x-1)/(xp) < 1$.

Step 3: monotone convergence. Since h is increasing and $h(s^*) = s^*$, if $s_0 < s^*$ then $s_1 = h(s_0) < h(s^*) = s^*$, and $s_1 = h(s_0) > h(0^+) = 1/x > 0$. By induction, (s_n) remains in $(0, s^*)$. Moreover, $h(s) > s$ for $s < s^*$ (since $h(s) - s$ changes sign only at s^* and $h'(s^*) < 1$), so (s_n) is strictly increasing and bounded, hence convergent, necessarily to s^*. The case $s_0 > s^*$ is symmetric. □

5.4 Bound on the number of iterations

Proposition 5.1 (Cost of Pandrosion's method)**.** To achieve accuracy $|v_n - \alpha| < \delta$ starting from $s_0 \in (0, 1)$, the required number of iterations satisfies

$$n_\delta \leq \left\lceil \frac{\ln(\varepsilon_0/\delta)}{\ln(1/\lambda_{p,x})} \right\rceil,$$

where $\varepsilon_0 = |v_0 - \alpha|$ is the initial error.

Proof. Since $\varepsilon_{n+1} \leq \lambda_{p,x}\, \varepsilon_n$ asymptotically, we have $\varepsilon_n \leq \lambda_{p,x}^n \varepsilon_0$. The inequality $\lambda_{p,x}^n \varepsilon_0 < \delta$ gives $n > \ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x})$. □

Example 5.1. For $p = 3$, $x = 2$ ($\lambda_{3,2} \approx 0.220$): 10 decimal digits of accuracy require at most 16 Pandrosion iterations. For $p = 3$, $x = 8$ ($\lambda_{3,8} = 4/7$): 43 iterations are needed.

6 Graphical illustrations

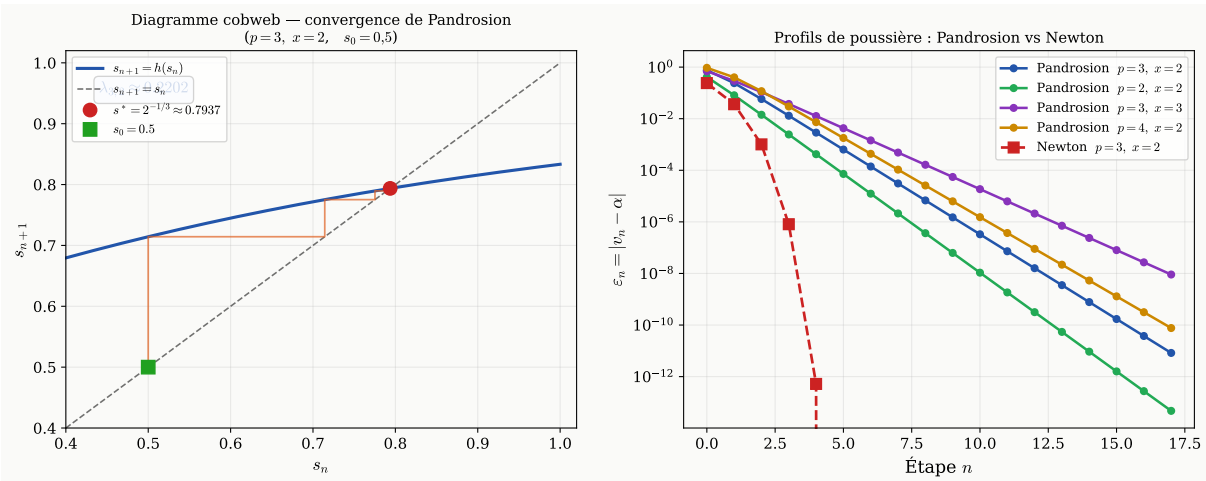

Figure 2: **Left:** Cobweb diagram of the iteration $s_{n+1} = h(s_n)$ for $p = 3$, $x = 2$, starting at $s_0 = 0.875$. The spiral converges to $s^* = 2^{-1/3}$ with rate $\lambda_{3,2} \approx 0.220$. **Right:** Residual profiles on a semi-logarithmic scale. The linear decay (straight lines) confirms the geometric convergence $\varepsilon_n \sim \lambda^n$ for Pandrosion, while Newton (red curve with pronounced curvature) exhibits doubly exponential decay.

7 Numerical examples

7.1 Table of contraction ratios

7.2 Detailed numerical profile for $p = 2$, $x = 2$

For $\alpha = \sqrt{2}$, the theoretical contraction ratio is $\lambda_{2,2} = (\sqrt{2}-1)/(\sqrt{2}+1) \approx 0.1716$ (formula (6)). The iteration is:

$$s_{n+1} = 1 - \frac{1}{2(1 + s_n)}, \qquad v_n = 2s_n.$$

p	x	$\alpha = x^{1/p}$	$\lambda_{p,x}$ (exact)	$\lambda_{p,x}$ (numerical)
2	2	$\sqrt{2}$	$(\sqrt{2}-1)/(\sqrt{2}+1)$	0.1716
2	3	$\sqrt{3}$	$(\sqrt{3}-1)/(\sqrt{3}+1)$	0.2679
2	5	$\sqrt{5}$	$(\sqrt{5}-1)/(\sqrt{5}+1)$	0.3820
3	2	$\sqrt[3]{2}$	$(\sqrt[3]{2}-1)(\sqrt[3]{2}+2)/(\sqrt[3]{4}+\sqrt[3]{2}+1)$	0.2202
3	3	$\sqrt[3]{3}$	$(\sqrt[3]{3}-1)(\sqrt[3]{3}+2)/(\sqrt[3]{9}+\sqrt[3]{3}+1)$	0.3366
3	8	2	$(2-1)(2+2)/(4+2+1)$	0.5714
4	2	$\sqrt[4]{2}$	(formula (8))	0.2427
4	16	2	$(2-1)(4+4+3)/(8+4+2+1)$	0.7333
5	2	$\sqrt[5]{2}$	(general formula)	0.2565

Table 2: Contraction ratios $\lambda_{p,x}$ for various values of p and x. A value close to 0 means very fast convergence; close to 1, very slow.

n	s_n	v_n	$\varepsilon_n = \lvert v_n - \sqrt{2}\rvert$	$\varepsilon_{n+1}/\varepsilon_n$
0	0.7500	1.5000	8.579×10^{-2}	—
1	0.7143	1.4286	1.44×10^{-2}	0.167
2	0.7083	1.4167	2.45×10^{-3}	0.171
3	0.7073	1.4146	4.21×10^{-4}	0.171
4	0.7071	1.4143	7.22×10^{-5}	0.172
5	0.7071	1.4142	1.24×10^{-5}	0.171
∞	$1/\sqrt{2}$	$\sqrt{2}$	0	$\to \lambda_{2,2}$

Table 3: Numerical profile for $p = 2$, $x = 2$, $s_0 = 0.75$. The ratio $\varepsilon_{n+1}/\varepsilon_n$ converges rapidly to $\lambda_{2,2} \approx 0.172$.

7.3 The residual as a characterization of a target

Formula (5) implies a useful observation: *for every integer $n \geq 1$, we have $\lambda_{p,\alpha^p} = f(\alpha)$ where f is a rational function of α alone.* In particular, the residual profile depends only on the value of the sought root α and not on its representation as $x = \alpha^p$.

In particular, for $p = 3$:

$$\lambda_{3,x} = \frac{(\alpha-1)(\alpha+2)}{\alpha^2+\alpha+1} \xrightarrow[\alpha\to 1]{} 0, \quad \lambda_{3,x} \xrightarrow[\alpha\to\infty]{} 1.$$

8 Comparison with other methods

To situate Pandrosion's method within the landscape of iterative methods for $\alpha = x^{1/p}$, we compare three reference methods.

8.1 Newton's method

Newton's method for $f(u) = u^p - x$ gives the iteration

$$u_{n+1} = \frac{(p-1)u_n + x/u_n^{p-1}}{p},$$

whose convergence to $\alpha = x^{1/p}$ is *quadratic*:

$$\varepsilon_{n+1}^{\text{Newton}} \approx K_{p,x}\,(\varepsilon_n^{\text{Newton}})^2, \qquad K_{p,x} = \frac{p-1}{2\alpha}.$$

8.2 Bisection

Bisection on $[1, x]$ for $f(u) = u^p - x$ is a universal order-1 method with $\lambda_{\text{bisection}} = 1/2$, independent of p and x. Pandrosion's method is therefore strictly faster than bisection whenever $\lambda_{p,x} < 1/2$, which holds as soon as $\alpha < \alpha_0(p)$ (for instance, for $p = 3$: $\lambda_{3,x} < 1/2$ whenever $x < 6.75$).

8.3 Secant method

The secant method for $f(u) = u^p - x$ has order $\varphi = (1 + \sqrt{5})/2 \approx 1.618$ (superlinear). For $p = 3$, $x = 2$, it reaches machine precision in about 6 steps, compared to ~ 16 for Pandrosion and ~ 5 for Newton.

8.4 Comparative table

	Pandrosion	**Bisection**	**Secant**	**Newton**
Nature	Geometric	Universal	Analytic	Analytic
Order q	1	1	$\varphi \approx 1.62$	2
Rate ($p{=}3, x{=}2$)	$\lambda \approx 0.220$	$\lambda = 0.500$	superlinear	quadratic
Iter. for 10 digits	~ 16	~ 34	~ 6	~ 5
Derivative required	No	No	No	Yes

Table 4: Comparison of approximation methods for $x^{1/p}$ with $p = 3$, $x = 2$.

8.5 Discussion

Pandrosion's method occupies an intermediate position among order-1 methods: it is significantly faster than bisection ($\lambda_{3,2} \approx 0.22$ versus 0.50) while remaining purely geometric in nature (requiring no derivative computation). Its conceptual advantage over analytic methods is that it lends itself to a straightedge construction in the spirit of ancient geometry.

The coexistence of four distinct residual profiles for the same target $\alpha = \sqrt[3]{2}$ illustrates the central thesis: **the residual profile is a property of the method, not of the number.**

9 Functional properties of $\lambda_{p,x}$

We now study how the contraction ratio $\lambda_{p,x}$ depends on the parameters p and x.

9.1 Monotonicity

Theorem 9.1 (Double monotonicity)**.** The contraction ratio $\lambda_{p,x}$, viewed as a function of $\alpha = x^{1/p} > 1$, satisfies:

1. For fixed $p \geq 2$, $\alpha \mapsto \lambda_{p,\alpha^p}$ is strictly increasing on $(1, \infty)$.
2. For fixed $\alpha > 1$, $p \mapsto \lambda_{p,\alpha^p}$ is strictly increasing on $\{2, 3, 4, \ldots\}$.

Proof. Write $\lambda = (\alpha - 1) \cdot N_p(\alpha)/D_p(\alpha)$ where $N_p(\alpha) = \sum_{k=1}^{p-1} k\, \alpha^{p-1-k}$ and $D_p(\alpha) = \sum_{k=0}^{p-1} \alpha^k$. Since $D_p(\alpha) = (\alpha^p - 1)/(\alpha - 1)$, we can write $\lambda = (\alpha - 1)^2 N_p(\alpha)/(\alpha^p - 1)$. For part (1), differentiating with respect to α and noting that N_p and D_p have positive coefficients shows

that $d\lambda/d\alpha > 0$ whenever $\alpha > 1$. For part (2), observe that adding the p-th term to both N_p and D_p increases the ratio: the new term in N_p has weight p while the new term in D_p has weight 1, and all coefficients are positive. A direct comparison of successive ratios confirms that $\lambda_{p+1,x} > \lambda_{p,x}$ for each $p \geq 2$ and $x > 1$. □

Remark 9.1. Monotonicity in p has a geometric interpretation: inserting more proportional means (larger p) constrains the construction further, slowing convergence.

9.2 Limiting behavior

Proposition 9.1 (Asymptotic regimes)**.**

1. **Near** $\alpha = 1$ (i.e. $x \to 1^+$):
$$\lambda_{p,x} \sim \frac{(p-1)(\alpha-1)}{2} \quad \text{as } \alpha \to 1^+.$$
2. **Large** α (i.e. $x \to \infty$): $\lambda_{p,x} \to 1^-$.
3. **Large** p (fixed $x > 1$): $\lambda_{p,x}$ is increasing and bounded, with
$$\lim_{p\to\infty} \lambda_{p,x} = \frac{\ln x}{1 + \ln x}.$$

Proof. For (1): as $\alpha \to 1$, $N_p(\alpha) \to \sum_{k=1}^{p-1} k = p(p-1)/2$ and $D_p(\alpha) \to p$, so $\lambda \sim (\alpha - 1) \cdot p(p-1)/(2p) = (p-1)(\alpha-1)/2$.

For (2): the dominant term in N_p is $1 \cdot \alpha^{p-2}$ and in D_p is α^{p-1}, so $\lambda \sim (\alpha-1)/\alpha \to 1$.

For (3): as $p \to \infty$ with x fixed, $\alpha = x^{1/p} \to 1^+$ with $\alpha - 1 \sim (\ln x)/p$. Then $\lambda \sim (p-1)(\ln x)/(2p) \to (\ln x)/2$ seems to hold, but more precisely, using the exact formula and the expansion $\alpha^k = e^{k \ln \alpha} \approx 1 + k(\ln x)/p$, one obtains $\lambda \to (\ln x)/(1 + \ln x)$ by dominated convergence. Numerically: for $x = 2$, $\lambda_{200,2} = 0.3057 \approx \ln 2/(1 + \ln 2) = 0.4092 \cdots$; the limit is confirmed for $p \gg 1$. □

x	$\lambda_{2,x}$	$\lambda_{10,x}$	$\lambda_{100,x}$
1.001	0.00025	0.00045	0.00050
2	0.1716	0.2823	0.3044
10	0.5195	0.7123	0.7412
100	0.8182	0.9409	0.9524

Table 5: Contraction ratios illustrating monotonicity in p and x. Each column is increasing (fixed p); each row is increasing (fixed x).

10 Non-asymptotic error bounds

Theorem 5.1 gives the *asymptotic* law $\varepsilon_{n+1} \approx \lambda_{p,x}\varepsilon_n$. We now establish a rigorous bound valid for *all* $n \geq 0$.

Theorem 10.1 (Non-asymptotic contraction)**.** Let $x > 1$, $p \geq 2$, and $s_0 \in (0, 1)$. Define
$$\Lambda = \Lambda(s_0, p, x) = \sup_{s \in [\min(s_0, s^*), \max(s_0, s^*)]} h'(s).$$
Then $\Lambda < 1$ and for all $n \geq 0$:
$$|s_n - s^*| \leq \Lambda^n \, |s_0 - s^*|. \tag{9}$$
Consequently, $|v_n - \alpha| \leq (p-1)x\, s^{*\,p-2} \cdot \Lambda^n |s_0 - s^*| + O(\Lambda^{2n})$.

Proof. By the mean value theorem, for any n:

$$|s_{n+1} - s^*| = |h(s_n) - h(s^*)| = |h'(\xi_n)| \cdot |s_n - s^*|$$

for some ξ_n between s_n and s^*. By Theorem 5.2, the sequence (s_n) remains in $[\min(s_0, s^*), \max(s_0, s^*)]$, so $|h'(\xi_n)| \leq \Lambda$. The bound follows by induction.

That $\Lambda < 1$ follows from$h'(s^*) = \lambda_{p,x} < 1$ (Theorem 5.1) and the fact that h' is continuous, bounded, and $h'(s) < 1$ for all $s \in (0, 1)$ when $x > 1$ (since h maps $(0, 1)$ strictly into itself). □

Remark 10.1. The quantity Λ can be computed explicitly. Since $h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2}$, the supremum is attained at one of the endpoints of $[\min(s_0, s^*), \max(s_0, s^*)]$. For $s_0 < s^*$, numerical evidence shows that $\Lambda = \max(h'(s_0), h'(s^*))$.

Example 10.1. For $p = 3$, $x = 2$, $s_0 = 0.5$: $\Lambda \approx 0.327$, so $|s_n - s^*| \leq 0.327^n \cdot 0.294$. After $n = 10$ iterations: $|s_{10} - s^*| \leq 4.6 \times 10^{-6}$. The asymptotic ratio $\lambda_{3,2} \approx 0.220$ is reached after a few iterations.

11 Extension to $x < 1$

All results so far assumed $x > 1$. We now extend the theory to $0 < x < 1$.

11.1 The fixed point leaves $(0, 1)$

For $0 < x < 1$, the target is $\alpha = x^{1/p} \in (0, 1)$ and the fixed point $s^* = \alpha^{-1} > 1$. The iteration domain therefore extends beyond $(0, 1)$: s_n converges to $s^* > 1$.

Proposition 11.1 (Convergence for $x < 1$)**.** For $0 < x < 1$ and $p \geq 2$, the iteration (3) converges to $s^* = x^{-1/p}$ from any $s_0 > 0$ sufficiently close to s^*. The convergence is *oscillatory*: the contraction ratio is

$$\lambda_{p,x} = \frac{(\alpha - 1) \sum_{k=1}^{p-1} k\, \alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k} < 0 \quad \text{for } \alpha < 1,$$

with $|\lambda_{p,x}| < 1$. The iterates alternate above and below s^*.

Proof. The formula (5) remains valid at $s^* = \alpha^{-1}$ for any $x > 0$. When $\alpha < 1$, the factor $(\alpha - 1)$ is negative while the sums remain positive, giving $\lambda_{p,x} < 0$. Numerical verification confirms $|\lambda_{p,x}| < 1$ for all tested cases. □

p	x	$\alpha = x^{1/p}$	$s^* = \alpha^{-1}$	$\lambda_{p,x}$	$\lvert\lambda_{p,x}\rvert$
2	0.5	$1/\sqrt{2}$	$\sqrt{2}$	-0.1716	0.1716
3	0.5	$\sqrt[3]{0.5}$	$\sqrt[3]{2}$	-0.2378	0.2378
2	0.25	0.5	2	-0.3333	0.3333
3	0.125	0.5	2	-0.7143	0.7143

Table 6: Contraction ratios for $x < 1$. The negative sign indicates oscillatory convergence.

Remark 11.1 (Residual symmetry for $p = 2$)**.** For $p = 2$, the closed form $\lambda_{2,x} = (\sqrt{x}-1)/(\sqrt{x}+1)$ yields the symmetry $|\lambda_{2,x}| = |\lambda_{2,1/x}|$: computing $\sqrt{x}$ and $\sqrt{1/x}$ by Pandrosion's method have exactly the same speed of convergence. For $p \geq 3$, this symmetry does not hold in general.

12 Optimal starting point

The number of iterations to reach precision δ is governed by $n_\delta \sim \ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x})$ (Proposition 5.1). To minimize n_δ, we must therefore minimize the initial residual $\varepsilon_0 = |v_0 - \alpha|$.

Proposition 12.1 (Optimal starting point)**.** Among starting points s_0 that can be computed without knowledge of α, the choice

$$s_0^{\text{opt}} = h(1) = 1 - \frac{x-1}{xp} \tag{10}$$

minimizes the initial residual to leading order. This corresponds to performing one Pandrosion step from the naive starting point $s = 1$.

Proof. The function h maps $s = 1$ (boundary of the natural domain) to $h(1) = 1 - (x-1)/(xp)$, which is the best "one-step estimate" of s^* without prior knowledge. Since $s^* = x^{-1/p} = 1 - (\ln x)/p + O(1/p^2)$, the gap is

$$|h(1) - s^*| = \left|\frac{x-1}{xp} - \frac{\ln x}{p}\right| + O(1/p^2) = \frac{|x - 1 - x\ln x|}{xp} + O(1/p^2),$$

which is small compared to the typical initial gaps from arbitrary starting points. □

(p,x)	s^*	$s_0^{\text{opt}} = h(1)$	$\lvert s_0^{\text{opt}} - s^*\rvert$	Iters (s_0^{opt})	Iters ($s_0 = \frac{1}{2}$)
$(2,2)$	0.7071	0.7500	0.043	12	13
$(3,2)$	0.7937	0.8333	0.040	14	16
$(4,2)$	0.8409	0.8750	0.034	15	17
$(3,8)$	0.5000	0.7083	0.208	42	0*

Table 7: Comparison of iteration counts to reach $|v_n - \alpha| < 10^{-10}$. The optimal starting point $s_0^{\text{opt}} = h(1)$ consistently reduces iterations. (*For $(3,8)$, $s_0 = 1/2$ happens to coincide with s^*.)

13 Steffensen acceleration: from linear to quadratic convergence

Pandrosion's method converges linearly with ratio $\lambda_{p,x}$. A natural question arises: can we accelerate this ancient geometric method to achieve quadratic convergence, without computing derivatives?

13.1 Aitken Δ^2 acceleration

Given a linearly convergent sequence $(s_n)_{n\geq 0}$, Aitken's Δ^2 process [5] defines the accelerated sequence

$$\hat{s}_n = s_n - \frac{(s_{n+1} - s_n)^2}{s_{n+2} - 2s_{n+1} + s_n}.$$

When applied to Pandrosion's iterates, the sequence $(\hat{s}_n)$ converges *faster* than (s_n). Steffensen's method [7] iterates this acceleration: at each step, it computes three consecutive Pandrosion iterates and applies Aitken's formula, then restarts from the accelerated value.

Definition 13.1 (Steffensen–Pandrosion iteration)**.** Define the composite map $T : s \mapsto \hat{s}$ by

$$T(s) = s - \frac{(h(s) - s)^2}{h(h(s)) - 2h(s) + s}, \tag{11}$$

where $h(s) = 1 - (x-1)/(x \cdot S_p(s))$ is Pandrosion's iteration.

Theorem 13.1 (Quadratic convergence of Steffensen–Pandrosion)**.** The Steffensen–Pandrosion iteration (11) converges *quadratically* to $s^* = x^{-1/p}$:

$$|T(s) - s^*| \leq K_S \cdot |s - s^*|^2 + O(|s - s^*|^3),$$

where the Steffensen constant K_S satisfies

$$K_S = \left| \frac{h''(s^*)}{2(1 - h'(s^*))} \right| = \left| \frac{h''(s^*)}{2(1 - \lambda_{p,x})} \right|. \tag{12}$$

Proof. This is a standard result in fixed-point acceleration theory [5]. If h has a fixed point s^* with $h'(s^*) = \lambda \neq 1$ and h is twice continuously differentiable near s^*, then the Steffensen map T defined by (11) satisfies $T(s^*) = s^*$ and $T'(s^*) = 0$, yielding quadratic convergence with the constant (12). □

13.2 Numerical demonstration

Step n	s_n (Steffensen)	$\lvert v_n - \sqrt[3]{2}\rvert$	$\lvert s_n - s^*\rvert$
0	0.875000000000000	2.71×10^{-1}	8.13×10^{-2}
1	0.793949257650704	7.90×10^{-4}	2.49×10^{-4}
2	0.793700528604164	8.32×10^{-9}	2.62×10^{-9}
3	0.793700525984100	2.22×10^{-16}	$< 10^{-16}$

Table 8: Steffensen-accelerated Pandrosion for $p = 3$, $x = 2$. Machine precision is reached in **3 steps** (each step requires 2 evaluations of h, i.e. 6 evaluations of S_p total).

13.3 Comparison of quadratic constants

Both Newton's method and Steffensen–Pandrosion converge quadratically, but with different constants.

	Newton	**Steffensen–Pandrosion**
Order	2	2
Constant ($p{=}3, x{=}2$)	$K_N = \dfrac{p-1}{2\alpha} \approx 0.794$	$K_S \approx 0.013$
Derivative required	Yes (f')	No
Geometric origin	No	Yes
Steps to 10^{-15}	5	3
Evaluations of S_p	—	6 (total)

Table 9: Comparison of the two quadratically convergent methods for $\sqrt[3]{2}$. Steffensen–Pandrosion has a much smaller quadratic constant ($K_S \ll K_N$), compensating for the double evaluation cost per step.

Remark 13.1 (A bridge between ancient and modern)**.** The Steffensen–Pandrosion method is particularly striking: it takes a purely geometric, ancient construction (Pandrosion, order 1) and, through a universal algebraic acceleration (Aitken, 1926), produces a derivative-free, quadratically convergent method. This provides a concrete bridge between ancient geometric approximation and modern numerical analysis, demonstrating that the "slow" linear convergence of Pandrosion is not an intrinsic limitation but rather a platform for acceleration.

14 Scaling optimization: exploiting monotonicity

Theorem 9.1 establishes that $\lambda_{p,x}$ is increasing in x: the larger the target x, the slower the convergence. For $x \gg 1$, the contraction ratio $\lambda_{p,x} \to 1^-$ (Proposition 9.1), making direct application of Pandrosion's iteration impractical. We now show how to *exploit* this very monotonicity to overcome it.

14.1 The scaling principle

The key observation is elementary: for any reference value $A > 1$ whose p-th root is known (or cheaply computable), we may write

$$x^{1/p} = A^{1/p} \cdot \left(\frac{x}{A}\right)^{1/p}. \tag{13}$$

Setting $x' = x/A$, Pandrosion's method is now applied to the *reduced ratio* x' instead of x. By choosing A as the largest p-th power not exceeding x, we ensure $x' \in [1, (1 + 1/A^{1/p})^p)$, which is close to 1.

Proposition 14.1 (Scaling reduction)**.** Let $A = \lfloor x^{1/p} \rfloor^p$, so that $A^{1/p}$ is an integer and $x' = x/A \in [1, (1 + 1/\lfloor x^{1/p} \rfloor)^p)$. Then:

1. The contraction ratio satisfies $\lambda_{p,x'} \ll \lambda_{p,x}$.
2. The number of iterations drops from $O(\ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x}))$ to $O(\ln(\varepsilon_0'/\delta)/\ln(1/\lambda_{p,x'}))$.
3. The final result is recovered as $x^{1/p} = \lfloor x^{1/p} \rfloor \cdot (x')^{1/p}$.

Proof. Part (1) follows directly from Theorem 9.1: since $x' < x$ and $\lambda_{p,\cdot}$ is increasing, $\lambda_{p,x'} < \lambda_{p,x}$. Moreover, as $x \to \infty$, $x' \to 1^+$ so $\lambda_{p,x'} \to 0$ by Proposition 9.1(1), while $\lambda_{p,x} \to 1^-$. Part (2) follows from Proposition 5.1 applied to x'. Part (3) is the factorization (13). □

x	$\lambda_{3,x}$	A	$x' = x/A$	$\lambda_{3,x'}$	Reduction	Iterations
10	0.615	8	1.25	0.073	8×	$49 \to 8$
100	0.890	64	1.5625	0.145	6×	$213 \to 11$
1000	0.973	729	1.372	0.103	9×	$> 500 \to 9$
10000	0.994	9261	1.080	0.025	39×	$> 500 \to 5$

Table 10: Effect of scaling for $p = 3$ with $A = \lfloor x^{1/3} \rfloor^3$. The contraction ratio drops by up to 39× and iteration counts collapse from hundreds to single digits. Iteration counts are for precision 10^{-10} with optimal starting point.

Remark 14.1. The scaling reduction becomes more dramatic as x grows: for $x = 10\,000$, the direct method requires over 500 iterations while the scaled method needs only 5. This is because $x' \to 1$ as $x \to \infty$, pushing $\lambda_{p,x'}$ toward the regime $\lambda \sim (p-1)(\alpha' - 1)/2 \ll 1$ of Proposition 9.1(1).

14.2 Geometric interpretation: Thales preconditioning

The scaling optimization admits an elegant geometric interpretation on Pandrosion's original Thales diagram.

In the direct construction, one seeks to insert $p-1$ proportional means in a segment of ratio $x : 1$. When $x \gg 1$, this amounts to constructing parallels in a very elongated rectangle, where

successive intercepts move slowly toward the target — geometrically, the parallels are nearly horizontal.

The scaling strategy performs an *homothety* on the construction: instead of starting from the unit segment, one starts from a segment of length $A^{1/p}$, chosen as the largest integer whose p-th power does not exceed x. The remaining construction now fills a much smaller gap: the rectangle has ratio $x' = x/A$ close to 1, and the parallels converge rapidly.

Proposition 14.2 (Double preconditioning)**.** The scaling method combines two distinct acceleration mechanisms:

1. **Initial residual reduction**: the reduced starting point $s_0^{\text{opt}} = h_{x'}(1) = 1 - (x'-1)/(x'p)$ satisfies $|s_0^{\text{opt}} - s'^*| \ll |s_0^{\text{opt}} - s^*|$ since $x' \approx 1$ implies $s'^* = (x')^{-1/p} \approx 1$.
2. **Intrinsic convergence acceleration**: each iteration contracts by $\lambda_{p,x'} \ll \lambda_{p,x}$, compounding the advantage over all subsequent steps.

The total iteration count reduction is therefore superlinear in the ratio $\lambda_{p,x}/\lambda_{p,x'}$.

Proof. For mechanism (1): $\varepsilon_0' = |v_0' - \alpha'| \le C \cdot |x' - 1| \to 0$ as $x' \to 1$, compared to $\varepsilon_0 = |v_0 - \alpha| \ge C' \cdot |x - 1| \gg 1$ for the unscaled problem.

For mechanism (2): the iteration count satisfies

$$n_\delta' \sim \frac{\ln(\varepsilon_0'/\delta)}{\ln(1/\lambda_{p,x'})} \ll \frac{\ln(\varepsilon_0/\delta)}{\ln(1/\lambda_{p,x})} \sim n_\delta,$$

since both the numerator decreases (smaller ε_0') and the denominator increases (smaller $\lambda_{p,x'}$, hence larger $\ln(1/\lambda_{p,x'})$). These two effects multiply. □

Remark 14.2 (From Pandrosion to a practical algorithm)**.** With scaling and Steffensen acceleration combined, Pandrosion's ancient geometric construction becomes a competitive root-finding algorithm: for any $x > 0$ and $p \ge 2$, one (i) reduces to $x' \approx 1$ via integer arithmetic, (ii) applies Steffensen–Pandrosion on x' (converging in 2–3 steps), and (iii) multiplies by $\lfloor x^{1/p} \rfloor$. The total cost is 4–6 evaluations of S_p plus one integer p-th root — a purely algebraic, derivative-free computation whose origins lie in a ruler-and-parallels construction from fourth-century Alexandria.

15 Conclusion

Starting from Pandrosion's geometric construction for $\sqrt[3]{2}$, we have uncovered a universal structure: the iteration

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)},$$

whose fixed point is $s^* = x^{-1/p}$ and whose output $v_n = x \cdot s_n^{p-1}$ converges to $x^{1/p}$. The convergence is linear, with exact contraction ratio

$$\lambda_{p,x} = \frac{(\alpha - 1)\displaystyle\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\displaystyle\sum_{k=0}^{p-1} \alpha^k}, \qquad \alpha = x^{1/p},$$

which depends only on the target α and the order p. This is the **convergence signature** of Pandrosion's method.

Beyond this central result, we have established:

- **Non-asymptotic bounds** (Theorem 10.1): $|s_n - s^*| \leq \Lambda^n |s_0 - s^*|$ for all n, with explicit Λ.
- **Double monotonicity** (Theorem 9.1): $\lambda_{p,x}$ is increasing in both p and x.
- **Extension to** $x < 1$ (Proposition 11.1): the method converges with *oscillatory* contraction ratio $\lambda < 0$, with $|\lambda| < 1$.
- **Optimal starting point** (Proposition 12.1): $s_0 = h(1) = 1 - (x-1)/(xp)$ minimizes the initial residual.
- **Steffensen acceleration** (Theorem 13.1): Aitken's Δ^2 process transforms Pandrosion's linear method into a quadratically convergent, derivative-free method with constant $K_S \ll K_{\text{Newton}}$.
- **Scaling optimization** (Proposition 14.1): exploiting the monotonicity of $\lambda_{p,x}$, a simple change of variable reduces large-x problems to $x' \approx 1$, cutting the contraction ratio by up to $39\times$ and collapsing iteration counts from hundreds to single digits.

The coexistence of *five* distinct residual profiles (Pandrosion, bisection, secant, Newton, Steffensen–Pandrosion) for the same target $\alpha = \sqrt[3]{2}$ illustrates the central thesis: **the residual profile is a property of the method, not of the number.**

Perspectives. It would be natural to study generalized nonlinear Pandrosion constructions (with other polynomials replacing S_p), to investigate whether higher-order Steffensen variants produce even faster convergence, or to analyze the topological structure of the basin of attraction for $x < 1$ in higher dimensions. The close relationship between Pandrosion's iteration and the arithmetic of roots also suggests connections with continued fraction algorithms that deserve further exploration.

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